

mFQS: A Multidimensional Visual Rating System

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Abstract

Expressing our opinions using rating systems in everyday life is a common practice. People often need to evaluate content for research, commercial applications, and personal expression. These evaluations provide important information for organizations. The limitation of current rating systems is that they often follow ordinal scales, like the Likert Scale, or are limited to binary voting. These rating systems only work on certain interfaces, lack flexibility to engage in nuanced evaluation requests, and can suffer from cultural biases. In this work, we introduce a graphical scale, the modified four-quadrant gradient rating scale (mFQS), designed to overcome these limitations. The mFQS allows users to rate multiple aspects at the same time while providing more granular control over the rating. The preliminary results show that the study participants found the system easier to use than the Likert scale and that there was a significant advantage in obtaining nuanced ratings from Japanese participants. Future directions for this work will look at implementing these scales in public contexts.

CCS Concepts

• **Human-centered computing** → HCI design and evaluation methods.

Keywords

Multidimensional Rating Systems; Music; Rating Behavior; Scales; Emotion; Human Computer Interaction; Human Factors

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1 Introduction

It is often necessary for industry and academic researchers to gather data on the qualitative observations [7, 21] and opinions [2, 12] of people. This information can be used to gather insights to build better products, understand people's perceptions, and see how viewpoints change [29, 30, 37]. Many rating approaches are limited in one major aspect; they use ordinal scales (e.g. Likert scales or star rating systems). Although these ordinal rating techniques are ubiquitous in society, they are not necessarily always the best method to use in every situation, as they suffer from presentation limitations on different devices and cultural bias [20, 22, 33, 34]. In this work, we develop a graphical multidimensional rating system called the modified four-quadrant gradient rating scale (mFQS), in which raters can give singular opinions on items that have two scales of interest. While the proposed system is straightforward, an initial experiment shows that the mFQS is preferable to the Likert scale for study participants and can even lead to less rating homogeneity with Japanese participants.

For HCI researchers, the Likert scale is likely familiar, as it is widely used to measure attitudes in survey questionnaires [14]. Originally designed as a balanced and symmetric 5-point scale, various lengths are possible, such as 7-point or 11-point scales [32]. Multiple guidelines exist for designing Likert scales, which researchers must carefully consider [31]. Capturing more complex topics with a Likert scale is difficult because conjunctions like "and", "or", or "but" should not be used [36]. Ratings on the Likert scale are susceptible to bias based on culture [13, 20, 33, 34] and frame of reference [26].

Only a handful of previous studies discuss the use of model rating scales. Most of these are kept one-dimensional [3, 6]. Therefore, there is an opportunity for capable rating systems that can be deployed across multiple device formats and contexts, freeing researchers from the limitations of current scales. An application area would be the analysis of political discourse. Social media recommendation algorithms overexpose users to content they like, creating a "social bubble" [24]. The mFQS can be used to rate news articles, marking possible bias by integrating it into a feedback loop [35]. However, the mFQS does not need to replace current systems, but can be used in combination with them. We aim to create a toolkit that combines existing systems with our novel scale, allowing for high customizability.

To test the mFQS (Figure 1), we used music as a domain. Music is a highly subjective and complex medium [4, 8, 15]. We compare the

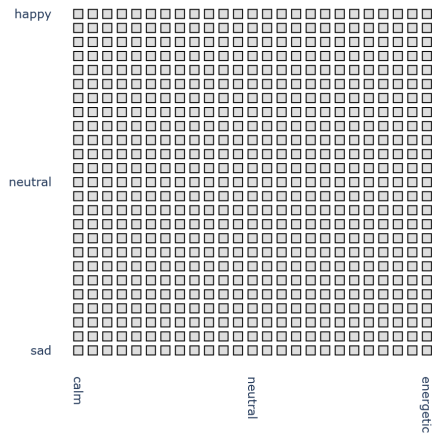


Figure 1: The modified Four Quadrant Scale (mFQS) labeled with "calm," "energetic," "sad," and "happy".

rating behavior on the mFQS with the Likert scale (Figure 2), the current gold standard. The mFQS' rating distribution was broader than that of the Likert scales, especially for the Japanese participants in the experiment, possibly alleviating cultural bias [13, 22, 26, 34]. There was no significant difference in rating time, indicating a similar cognitive load and learning curve. According to the post-hoc survey, participants preferred the interaction with the mFQS over the Likert scale because it was more intuitive and helped them visualize their feelings rather than having to choose a number.

Our research opens up new possibilities for more expressive rating interfaces, as ratings for highly subjective and personalized domains can be difficult for users to input on conventional scales. Our first exploration in a desktop setting with music lays a stepping stone for expanding the research of multidimensional scales to more devices, contexts, and domains.

2 Methodology

In our experiment, we focused exclusively on instrumental music, excluding lyrics, to make it more accessible to participants from diverse cultural backgrounds. The master playlist was randomly chosen from four different datasets with various genres of music: AAM (algorithmically created music) [27, 28], Babyslakh (a subset of Slakh2100, music in various genres) [17–19], EMOPIA (pop music piano covers) [10, 11], and MAESTRO (classical piano music) [9].

We randomly selected 60 musical clips from the datasets, 20 of which had pairs and 40 did not, totaling 80 music clips that each participant evaluated. We defined a pair as two music clips from the same song. Each clip was shortened to 20 seconds in length. We randomly assigned each clip to either the Likert scales or the mFQS, resulting in 39 clips on the Likert scales and 41 on the mFQS. This distribution is slightly offset because it was designed per dataset, disregarding the total distribution. For paired songs, we evenly distributed 20 clips for each scale type. The participants rated the same set of clips with the same scale assignment. We randomized the playlist for each participant and introduced clips without pairs as "noise" to make it more difficult for them to identify this pattern.

Randomization was used to prevent the introduction of regularity patterns or learning effects.

Both scales were designed for direct comparison between rating results and have a granularity of 25 stops; each Likert scale had 25 rating points, and the mFQS was a 25×25 grid. The listener rated each clip on two attributes: how "calm or energetic" and "sad or happy" each clip felt. The combinations of varying degrees of "calm and sad," "calm and happy," "energetic and sad," and "energetic and happy" are possible, which gives us the four quadrants of the mFQS. The Likert scale interface (Figure 2) displays two different Likert scales, while the mFQS interface (Figure 1) displays one mFQS.

3 Experiment

We conducted an experiment with 20 participants (16 male, 4 female). The age range was between 19 and 32 years, with a mean of 23.65 years and a standard deviation of 3.27 years. The nationalities included ten Japanese and ten students from seven other countries. All participants were undergraduate or graduate students majoring in various technology-based fields. Recruitment was carried out through university message boards. The experiment was conducted in English and Japanese. This work received ethical approval from the institution's ethics committee prior to commencement.

Due to the sensitive nature of music, each experiment was conducted in a controlled room with the same setup to maintain consistency and minimize outside influences. Bose Noise Canceling Headphones 700 were used to block external noise. All experiment sessions lasted 1 hour and 30 minutes. The total listening and rating time was 1 hour with a 15-minute break in the middle. During this break, participants were asked not to eat, drink caffeinated beverages, or smoke. Non-caffeinated drinks were allowed. This restriction was implemented to ensure uniformity among all participants in the experiment as best as possible. Each participant listened to a total of eighty 20-second music clips with a break after 40 clips. Before starting the experiment, participants signed a consent form. Afterward, they completed a post-hoc survey and received a remuneration of 1500 JPY (approximately 10 USD).

The participants first selected between English and Japanese on the interface. They then received a textual explanation of the experiment flow and the scales before starting to evaluate clips. For each clip, the music started when the participant pressed the play button. The scales were not displayed during listening. Participants could not stop or replay clips. It was possible to mute a clip if the music caused any distress. After 20 seconds, the music stopped automatically and one of the rating scales was displayed. The participants then had 25 seconds to rate their feelings about the clip played. If this timer ran out, the interface skipped automatically to the next clip. Participants could also skip to the next clip by themselves. However, participants cannot return to a previous clip, re-listen, or change their ratings after loading the next page.

4 Results

We analyzed the ratings of song pairs. Both scales are designed with the same granularity of 25 stops and a 25×25 grid to allow a direct comparison of the ratings on both scales.

To compare the rating distribution, we calculated the mean, median, and standard deviation per axis. In Table 1, we can observe

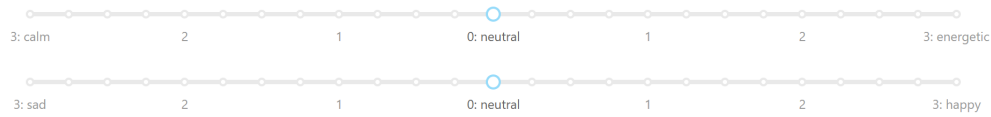


Figure 2: The Likert scales, where the upper scale is labeled "calm" and "energetic" and the lower is labeled "sad" and "happy."

Table 1: Rating Distribution over "Calm/Energetic"-Axis

	Mean	Median	Standard Deviation
mFQS	0.193	0.50	1.589
Likert	-0.038	0.25	1.467

Table 2: Rating Distribution over "Sad/Happy"-Axis

	Mean	Median	Standard Deviation
mFQS	-0.103	0.000	1.509
Likert	-0.088	-0.125	1.250

that the mean and median of the "Calm/Energetic"-axis is higher on the mFQS compared to the Likert scale. Using the Wilcoxon test, this difference is statistically significant (p-value: 0.00223). In Table 2, we can see that on the "Sad/Happy"-axis, the mean and median are very close, with the mFQS values being slightly higher. This difference is not statistically significant (p-value: 0.35193).

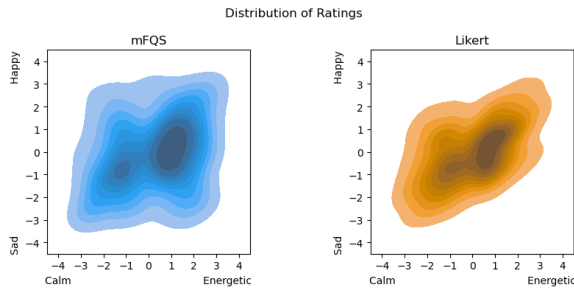


Figure 3: Distribution of ratings for the mFQS and the Likert scales mapped into a 2D coordinate system.

In Figure 3, we directly map the two Likert scales to a 2D coordinate system to analyze the ratings on both scales. We can observe that the Likert scales' rating distribution is narrower than that of the mFQS. The Likert scales' rating has a more linear correlation between the two axes, whereas the mFQS' distribution is much broader across the entire space. The two clusters for both rating scales lie in similar areas. Participants tend to be more expressive on the "Calm/Energetic"-axis using the mFQS compared to the Likert scale. The distribution shows two peaks with higher ratings at the extreme ends of the scale for the mFQS. The "Sad/Happy"-axis distribution only shows one peak. Participants still tend to choose more ratings between $[-1, 1]$ on the Likert scale and rate more on the extreme ends of the mFQS.

Next, we tested for differences in the rating behavior of nationalities. We divided our participants into two groups: Japanese (10

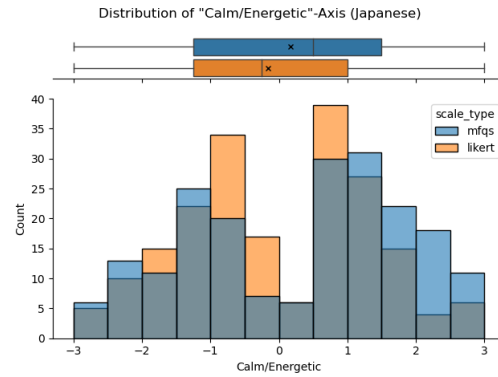


Figure 4: Rating distribution of "Calm/Energetic"-axis in a histogram for Japanese participants.

participants) and non-Japanese (10 participants). We compared the rating distribution for each axis on both scales within their respective nationality groups. The "Calm/Energetic"-axis showed a statistically significant difference using the Wilcoxon test for the Japanese group (p-value: 0.002230). The Likert scale's ratings are much closer to the neutral point, whereas the mFQS is used more toward the extreme ends (Figure 4). Previous research shows that Japanese people tend to rate closer to the neutral point on conventional Likert and ordinal scales [13, 22]. The distribution of the mFQS could indicate a reduction of cultural bias.

To test for cognitive load, we also analyzed the time it took participants to rate on either scale. The average interaction time on the mFQS is slightly higher than that for the Likert scales. The difference is not statistically significant using the Wilcoxon test (p-value: 0.28797). In addition, we investigated whether there was a difference in the learning effect when using the scale. The interaction time decreased over time for both scales with a similar trend, which implies no difference in learning behavior. The observation is highlighted by a non-statistically significant Wilcoxon test result (p-value: 0.18815).

In the post-hoc survey, participants were asked to rate the user interaction using the Likert scales and the mFQS on a scale from 0 (very difficult to use) to 10 (very easy to use) and to indicate which one they preferred. The score of the mFQS was, on average, slightly higher at 6.9, compared to the Likert scale at 6.5; however, the difference is not statistically significant. In response to the question "Which one did you prefer and why?" twelve participants preferred the mFQS, and eight preferred the Likert scale. Many participants chose the mFQS because the scale is easier and more intuitive to use, allowing them to easily imagine a rating and visualize their feelings. On the other hand, eight participants preferred the Likert

scale because it allows for easy comparison between axes if the ratings go in the same direction, for example, if a song feels calmer than sad.

5 Discussion

Rating online content, products, or services is often encountered in daily life as a ubiquitous task [2, 7, 12, 21]. Typical ordinal scales are commonly used for these tasks, but are limited by design. Our study introduced a novel multidimensional scale for assessing emotions, called mFQS. We compared the rating behavior on the Likert scale and the mFQS and found that the mFQS provides a compelling alternative method to obtain qualitative observations from individuals.

Our preliminary exploration into the mFQS opens up new possibilities for further research. The goal should not be to replace the current methods outright, but to enhance certain domains with more nuanced feedback collection tools, including those that use spatial visual design. Our exploration in music demonstrated that the mFQS is effective. We aim to create a toolkit that allows other researchers to quickly adjust the scale to their specific needs. In domains where greater expressiveness is desirable, such as medicine, pain management, depression, and highly personal topics, the mFQS could be an excellent tool for self-assessment. Due to the ease of use, it could also be employed on multiple device form factors, opening up the rating system for in-the-wild use. Through the more granular selection options, it is easier to select a rating compared to ordinal scales with pre-defined rating positions, especially on mobile devices.

There are limitations to this study. To validate a new rating system, the number of participants is relatively small. In addition, our participants are from a selected demographic of university students, with females being underrepresented and a limited age distribution. To demonstrate that the scale works for the broader population, we need to conduct more rigorous tests for reliability, validity, and sensitivity [5]. Furthermore, the experiment was conducted in a controlled environment using a desktop setup. An in-the-wild study using different devices could alter rating behavior. Portability to other domains should also be explored. Our current exploration is limited to the domain of music.

6 Future Work

Our analysis found significant differences in the usage of the mFQS and the Likert scale. We consider the mFQS an intriguing rating interface that should be explored further. We aim to make the mFQS available in areas where one-dimensional scales may only partially capture certain aspects of an individual's opinion. The mFQS offers the necessary customization and multidimensionality to ask more complex questions. Politics, education, and medicine are all domains that require the evaluation of complex and interconnected topics. The multidimensionality of the mFQS gives a visual tool for more nuanced ratings. Combined with current systems such as the Likert scale, it provides a potential toolkit for evaluating human-centric topics. For easy accessibility, we aim to create a toolkit that allows researchers to customize the scale to their specific needs, such as adjusting the granularity, adjectives, sensor integration, and adaptability to various device form factors. Figure 5 shows another

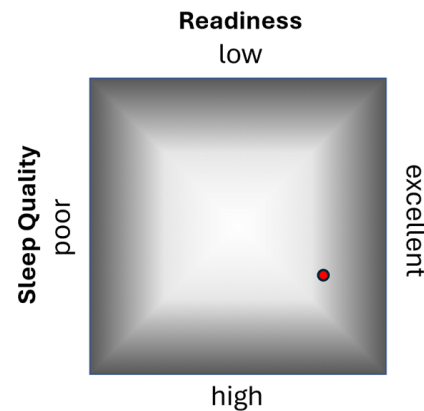


Figure 5: An updated design of the modified Four Quadrant Scale (mFQS) labeled with "poor," "excellent," "low," and "high" in the context of rating sleep.

representation with higher granularity, making it better suited for mobile devices.

In our current research, we are deploying the mFQS in-the-wild for self-assessment in sleep. Wearable smart rings and watches can classify and evaluate sleep quality [1, 16]. However, individual perception might disagree with their sleep score [25]. The mFQS can be used for self-assessment on a smartphone to collect subjective evaluation, for example, to connect factors such as sleep quality and readiness (Figure 5). From previous research, we know that the perceived self-evaluated rating in combination with the measured variable from the ring gives strong predictors of working memory [23] that are better indicators than sleep itself. However, collecting this information with ordinal scales is a burdensome task. Using a tool like the mFQS can help us to directly connect two highly correlated dimensions, such as quality of sleep and readiness (cognitive and / or physical) in a visual manner, giving a new option for obtaining feedback from study participants.

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